

Charge Densities in Real Space: RealQM and the Realist Quest Carried by Schrödinger

Author. Claes Johnson

Affiliation. KTH Royal Institute of Technology, SE-100 44 Stockholm, Sweden

Corresponding author. Claes Johnson, claesjohnson@gmail.com

Homepage. <https://claes542.github.io/RealMolecule/gallery.html>

Declarations

Funding. None.

Competing interests. The author declares no competing interests.

Data availability. The computational model and the interactive simulations referred to in the manuscript are openly available at the author’s public repository (<https://github.com/Claes542/RealMolecule>) and gallery (<https://claes542.github.io/RealMolecule/gallery.html>). No datasets were generated for this article.

Acknowledgements. None.

Note on the anonymized references. Several bibliography items are the author’s own companion work; in the blinded manuscript their author name and identifying URLs are withheld for double-anonymous review (shown there as “[Author]” with details withheld). Their full forms are:

- C. Johnson, *Quantum Mechanics as Multiphase 3D Continuum Mechanics*, manuscript, 2026; <https://claes542.github.io/RealMolecule/gallery.html>.
- C. Johnson, *A 3D Multiphase Continuum Computational Model for Atoms*, manuscript, 2026.
- C. Johnson, *Nuclear Binding as Dual Coulomb Confinement, without a Strong or Weak Force*, manuscript, 2026.
- C. Johnson, *The Binding Energy Problem QCD Never Solved — and a Coulomb Model That Does*, manuscript, 2026.
- C. Johnson, *Solar Fusion by Coulomb Alone*, manuscript, 2026.
- C. Johnson, *RealQM vs the Standard Model: Extension or Point, One Term or Many*, manuscript, 2026.
- C. Johnson, *The RealQM Gallery: interactive charge-density simulations of atoms, molecules, reactions, and nuclei*, 2026; <https://claes542.github.io/RealMolecule/gallery.html> (source: <https://github.com/Claes542/RealMolecule>).

Claes Johnson
KTH Royal Institute of Technology
Stockholm, Sweden
claesjohnson@gmail.com

To the Editors-in-Chief, *Synthese*

Dear Editors,

I submit for your consideration the manuscript “Charge Densities in Real Space: RealQM and the Realist Quest Carried by Schrödinger” for publication in *Synthese*.

The paper takes up a question in the history and philosophy of quantum mechanics — whether the microscopic world is of the same form as the macroscopic, extended matter in three-dimensional space only of smaller size, or something categorically other — and connects it to two current debates usually treated separately: (i) the *arena* of the wave function, wave-function realism (Albert, Ney) versus the primitive-ontology programme (Maudlin, Allori, Dürr–Goldstein–Zanghì, Esfeld), with spacetime state realism (Wallace & Timpson) as a middle route; and (ii) the *individuality* of identical particles, whether permutation-invariant particles are individuals and whether the Identity of Indiscernibles survives (French & Krause; Saunders; Muller & Saunders; Muller & Seevinck; Caulton & Butterfield).

The central observation is that both problems are consequences of two features of one formula — the antisymmetric N -particle amplitude on $3N$ -dimensional configuration space: the high-dimensional domain generates the arena problem, and the antisymmetry generates the individuality problem. I then examine a real-space charge-density formulation (“RealQM”), which revives Schrödinger’s original 1926 reading of the wave function as a charge density in ordinary space — the *anschaulich* programme displaced by the Bohr–Born–Heisenberg–Dirac turn precisely because, for many electrons, the amplitude moves onto $3N$ -dimensional configuration space. RealQM keeps N unit charge densities on non-overlapping regions of $\mathbb{R}^3$, and so has neither feature: it is a primitive ontology in three-space, and it individuates its electrons absolutely, by the regions they occupy, so that the Identity of Indiscernibles is satisfied. The two virtues are a single geometric fact — the non-overlap of the regions does double duty — and Pauli exclusion and the identity of indiscernibles coincide.

I give the argument in explicit formulas, on the view that here philosophy is made more precise with them than without. A substantial section argues that the decisive asset is computational: the $3N$ -dimensional amplitude is intractable beyond very small N — practice already retreats to effective three-dimensional objects, and density-functional theory makes the 3D density fundamental in practice — whereas RealQM computes directly with its beables in $\mathbb{R}^3$, and the same energy functional has been used to compute atoms across the periodic table, molecular bonds and hydrogen bonds, and, reading the nucleus as protons and electrons bound by Coulomb alone, nuclear binding energies. That one parameter-poor functional reproduces observation across five orders of magnitude in scale is offered, on no-miracles grounds, as evidence that the model tracks real structure. A section places the proposal against the point-particle ontology of the Standard Model; a further section meets the main objections; and a closing reflection reads the result as a Wittgensteinian dissolution of the two problems — not deflationary but a redirection toward the nature of physical reality itself. I am careful throughout to bound the claim: RealQM is a rival theory whose

warrant is its computed chemistry and nuclear structure, and the metaphysical verdict is explicitly conditional on that empirical adequacy.

The manuscript is original, is not under consideration elsewhere, and has a single author. In keeping with *Synthese*'s double-anonymous review policy, the manuscript is submitted blinded, with a separate title page (above) carrying the author information, declarations, and the withheld reference details.

Thank you for considering the manuscript.

Sincerely,
Claes Johnson